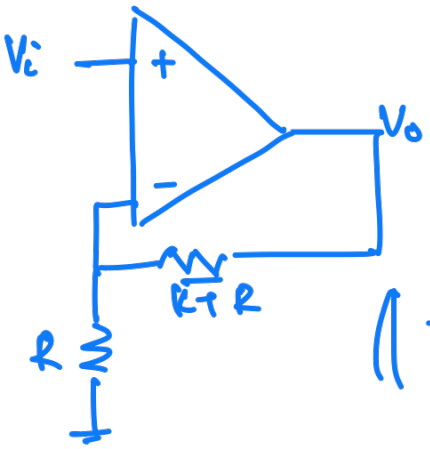
Lecture #12



$$A(s) = \frac{A_{02}}{\left(1+\frac{s}{\omega_{p1}}\right)\left(1+s/\omega_{p2}\right)}$$

$$\left(\frac{V_o}{V_i}\right) \frac{\frac{1}{k}}{1+\frac{1}{A_0}} \frac{1}{1+\frac{s\,\omega_{p1}+\omega_{p2}}{\left(1+\frac{A_{02}}{k}\right)\omega_{p1}\omega_{p2}} + \frac{s^2}{\left(1+\frac{A_{02}}{k}\right)\left(\omega_{p1}\omega_{p2}\right)}}$$

$$\frac{s}{\omega_p'} = \frac{-1}{2Q_p} \pm j\sqrt{1-\frac{1}{4Q_p^2}}$$

$$D(s) = 1+\frac{s\,\omega_{p1}+\omega_{p2}}{\left(1+\frac{A_{02}}{k}\right)\omega_{p1}\omega_{p2}} + \frac{s^2}{\left(1+\frac{A_{02}}{k}\right)\left(\omega_{p1}\omega_{p2}\right)}$$

$$= 1+\frac{s}{\omega_p' Q_p} + \frac{s^2}{\omega_p'^2}$$

$$\omega_p' = \sqrt{1+\frac{A_{02}}{k}}\ \sqrt{\omega_{p1}\omega_{p2}}$$

$Q_p \le \frac{1}{2}$, Poles are real

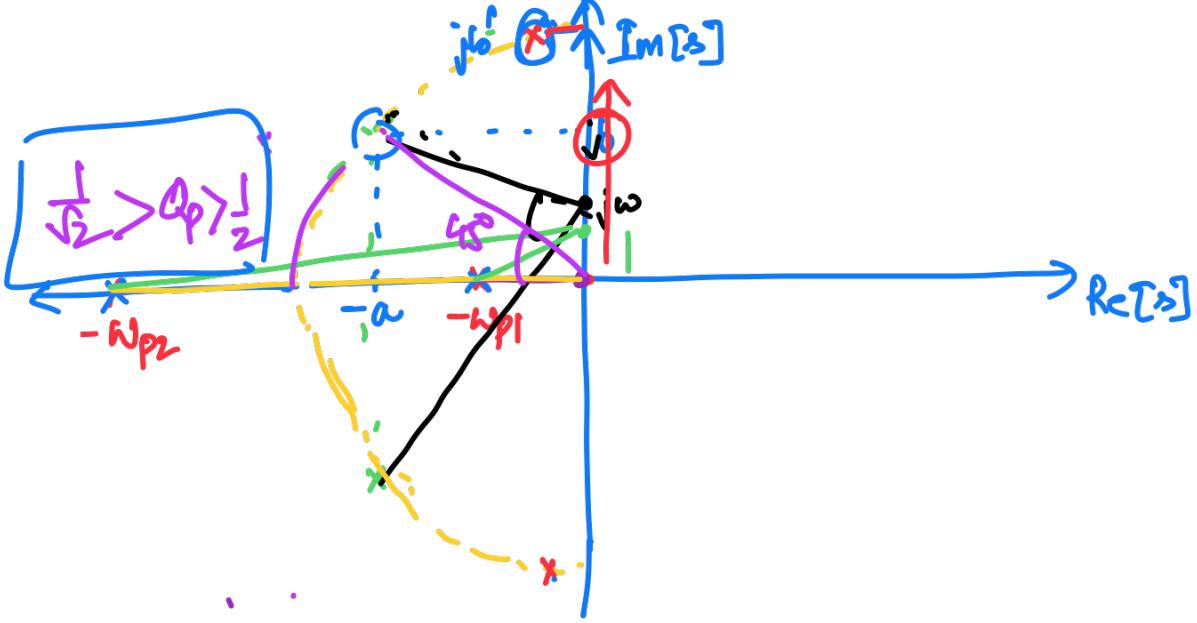
$Q_p > \frac{1}{2}$, Poles are complex

$$\omega_p' Q_p = \left(1+\frac{A_{02}}{k}\right)\frac{\omega_{p1}\omega_{p2}}{\omega_{p1}+\omega_{p2}} \Rightarrow Q_p = \sqrt{1+\frac{A_{02}}{k}}\ \frac{1}{\sqrt{\frac{\omega_{p1}}{\omega_{p2}}}+\sqrt{\frac{\omega_{p2}}{\omega_{p1}}}}$$

Case #1 $\omega_{p1} = \omega_{p2} \Rightarrow$ $Q_p = \dfrac{\sqrt{1+\frac{A_0}{k}}}{2} > \frac{1}{2}$

Case #2 $\omega_{p1} \gg \omega_{p2}$, $\omega_{p1} = 100\,\omega_{p2}$

$$Q_p = \frac{\sqrt{1+\frac{A_{02}}{k}}}{\sqrt{\frac{\omega_{p1}}{\omega_{p2}}}} = \frac{\sqrt{1+\frac{A_{02}}{k}}}{10}$$



$$\frac{V_o}{V_i} = \frac{k}{1 + k/A_{02}} \cdot \frac{1}{\left(1 + \frac{s}{p_1}\right)\left(1 + \frac{s}{p_2}\right)}$$

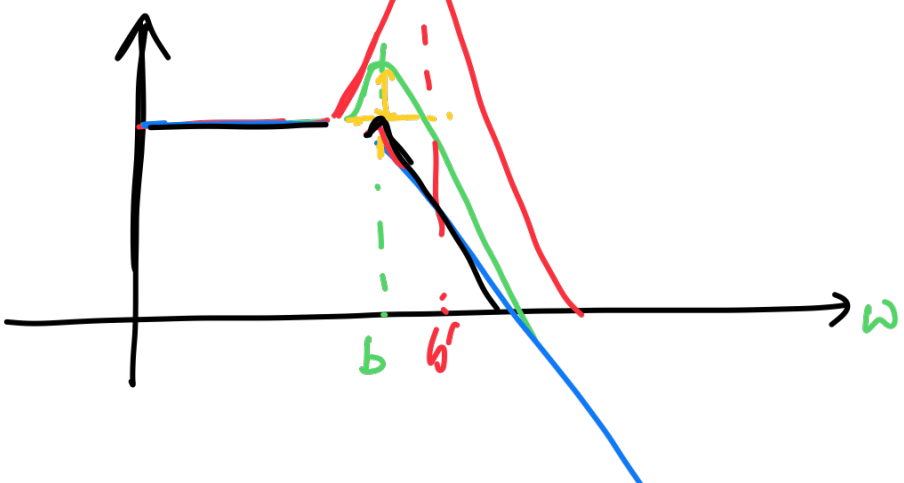
p_1, p_2 are roots for $D(s) = 0$

For complex poles,

$$\frac{V_o}{V_i} = k_{dc} \frac{1}{[s + (a + jb)][s + a - jb]} \quad ; \; a > 0$$

$$\left|\frac{V_o}{V_i}\right|_{s=j\omega} = k_{dc} \frac{1}{[j\omega + a + jb][j\omega + a - jb]}$$

$$|D(j\omega)|^2 = \cdot \left(a^2 + (b+\omega)^2\right)\left(a^2 + (b-\omega)^2\right)$$



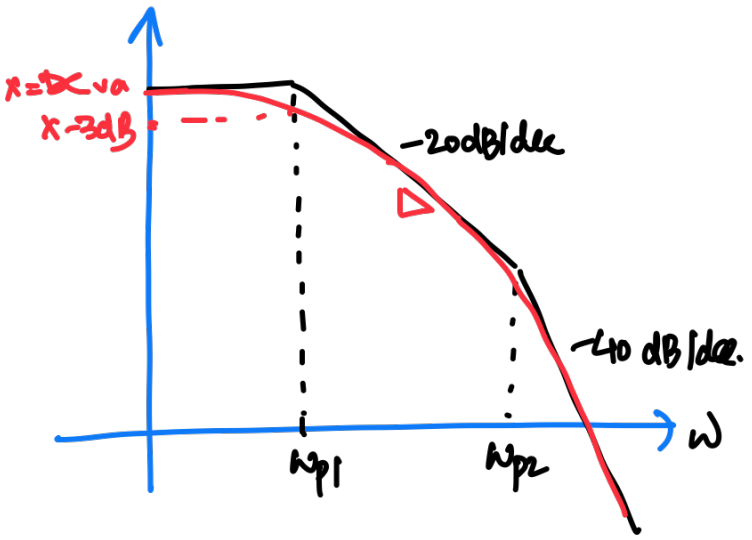
$$H(s) = \frac{A_{DC}}{\left(1+\frac{s}{\omega_{p1}}\right)\left(1+\frac{s}{\omega_{p2}}\right)}$$

$$H(0) = A_{DC}$$

$$|H(j\omega_{p1})| = \frac{A_{DC}}{\left|1+j\frac{\omega_{p1}}{\omega_{p1}}\right| \left|1+j\frac{\omega_{p1}}{\omega_{p2}}\right|}$$

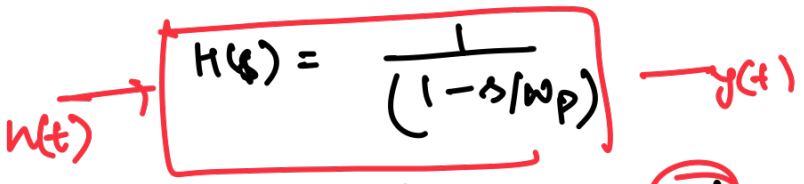
$$= \frac{A_{DC}/\sqrt{2}}{\left|1+j\frac{\omega_{p1}}{\omega_{p2}}\right|}$$

$$\simeq \frac{A_{DC}}{\sqrt{2}}$$



$20\log|\ | = A_{DC} - 3\,dB$

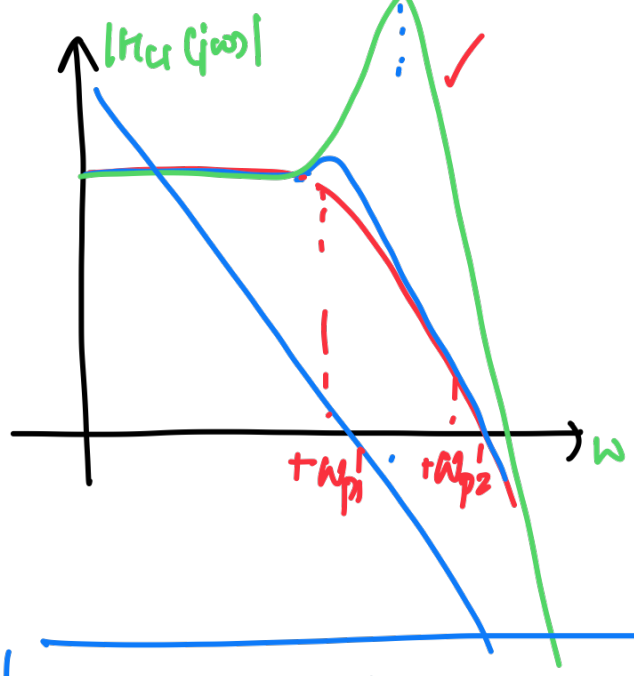
$-3dB$ Bandwidth.

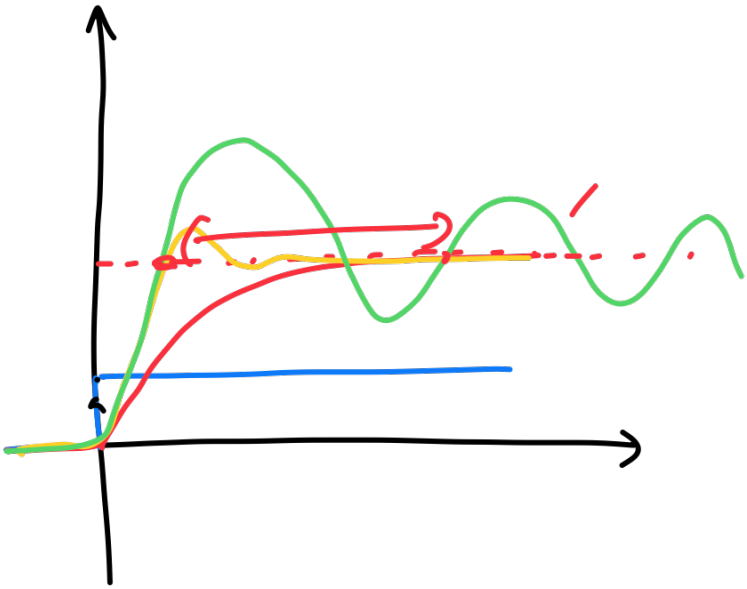


$$h(t) = \mathcal{L}^{-1}(H(s)) = e^{\omega_p t}$$

$$Y(s) = \frac{1}{s} \times \frac{-\omega_p}{s-\omega_p}$$

$$= \frac{1}{s} + \frac{-\omega_p}{s-\omega_p}$$





$$u \rightarrow \boxed{H_{cl}(s) = \frac{k_{dc}}{(s+a+jb)(s+a-jb)}} \rightarrow \text{(for complex poles)}$$

$$u(t) \longrightarrow u(s) = 1/s$$

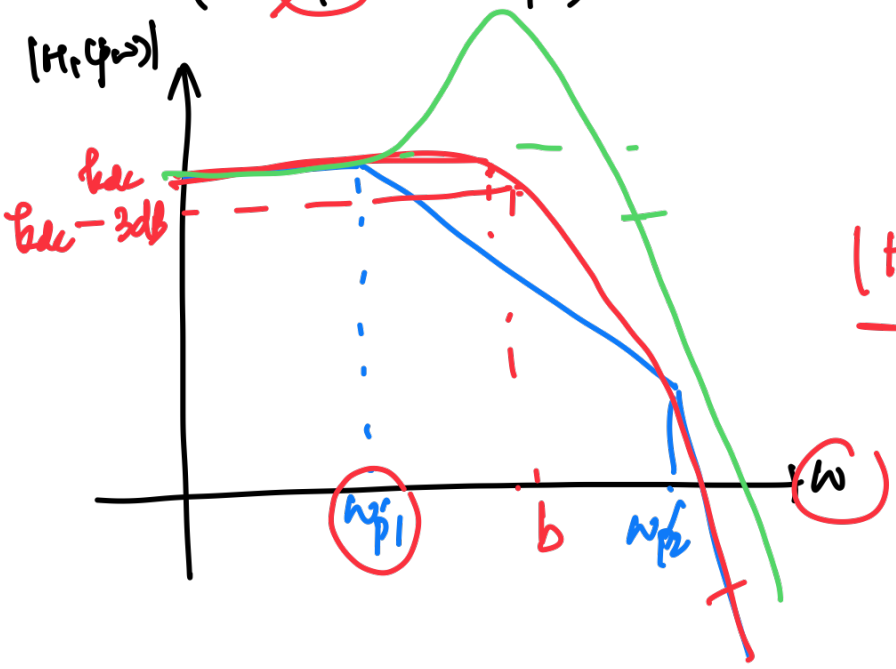
$$Y(s) = \frac{1}{s} \frac{k_{dc}}{(s+a+jb)(s+a-jb)}$$

$$= \frac{B}{s} + \frac{C+jD}{s+a+jb} + \frac{C-jD}{s+a-jb}$$

$$y(t) = \mathcal{L}^{-1} Y(s) = B\,u(t) + (C+jD)e^{-(+a+jb)t} + (C-jD)e^{-(a-jb)t}$$

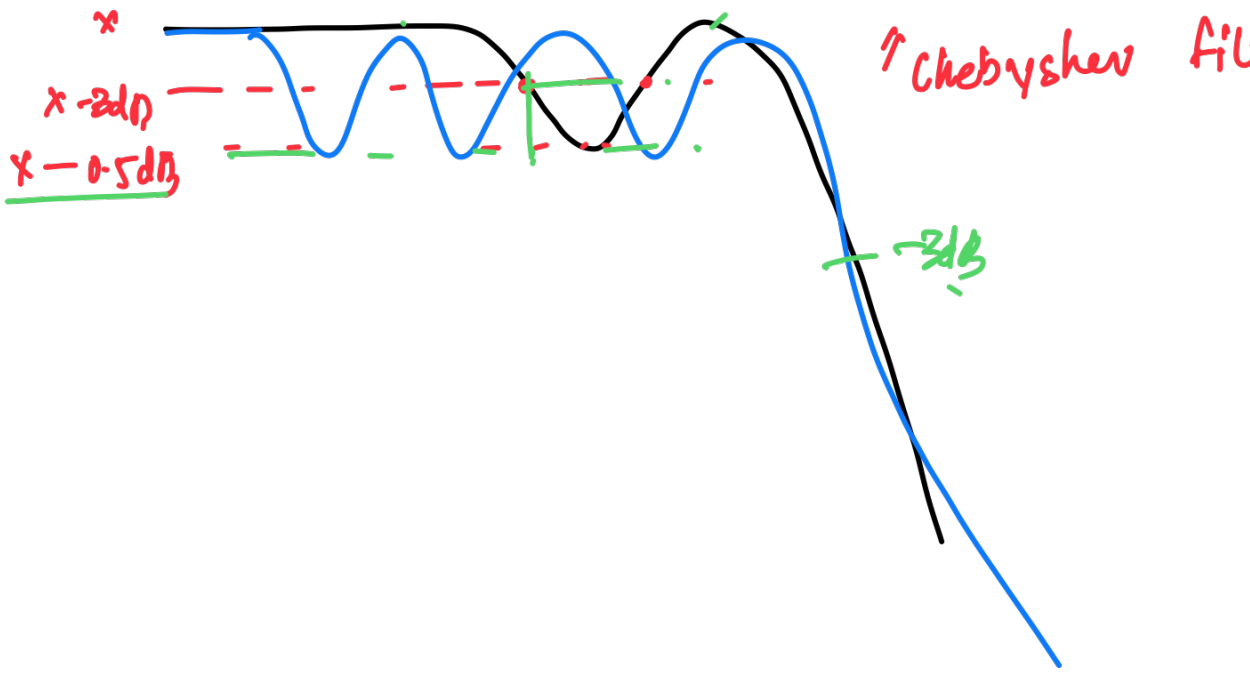
$$= B\,u(t) + C\left(e^{-(a+jb)t} + e^{-(a-jb)t}\right) + jD\left(e^{-(a+jb)t} - e^{-(a-jb)t}\right)$$

$$H_1(s) = \frac{k_{dc}}{\left(1+\frac{s}{\omega_{p1}'}\right)\left(1+\frac{s}{\omega_{p2}'}\right)} \quad ; \quad H_2(s) = \frac{k_{dc}}{\left(1+\frac{s}{a+jb}\right)\left(1+\frac{s}{a-jb}\right)}$$



$s = j\omega$

$$|H_2(j\omega)| = \frac{1}{\sqrt{2}}|H_2(j0)|$$

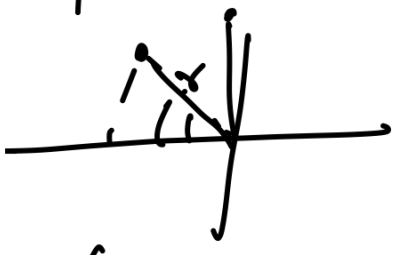


↑ Chebyshev filter"

$$D(s) = \frac{s^2}{\omega_p'^2} + \frac{s}{\omega_p' Q_p} + 1$$

Q_p: Quality factor.

$$\frac{s}{\omega_p'} = \frac{-1}{2Q_p} \pm j\sqrt{1 - \frac{1}{4Q_p^2}}$$



$$\gamma = \omega_p' \sqrt{\frac{1}{4Q_p^2} + 1 - \frac{1}{4Q_p^2}} = \omega_p'$$

$$D(s) = \left(s^2 + 2\zeta\omega_n s + \omega_n^2 \right)$$

ζ : Damping Coeff.

$$= \omega_n^2 \left(\frac{s^2}{\omega_n^2} + \frac{2\zeta}{\omega_n} s + 1 \right)$$

$$\frac{1^2}{\omega_p'^2} + \frac{s}{\omega_p' Q_p}$$

$$Q_p = \frac{1}{2\zeta}$$